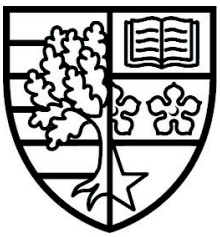
Skin Lesion Classification on FPGA

by Lewis Turner

Research Report

Submitted in partial fulfilment for the degree of
BSc (Hons) Computer Science

Supervised by Dr. Rob Stewart, Ms Tess Watt



Heriot-Watt University
School of Mathematical and Computer Sciences

6th November 2026

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Type of assessment	Individual
Document title	Skin Lesion Classification on FPGA
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• Google Gemini 3.8 Flash (Extended Thinking) to assist in searching for relevant research papers.

• Google Gemini 4.0 Argon (thinking: Low - High) to assist setting up FPGA for use over Wifi.

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Acknowledgements

Acknowledgements text.

Abstract

Abstract text.

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1 Introduction

Summary of project motivation & context

1.1 Background and Motivation

- Using papers (mainly) from literature review to provide background context
- Motivations for doing this project
- Identifying problems to be solved to achieve objectives

1.2 Aim and Objectives

State project aims
Table of objectives with IDs to reference later

2 Literature Review

Grouping topic summaries, critical assessment, gaps

2.1 Search Methods

I used Google Gemini 3.8 Flash on thinking mode High (October 2026 version) to source the most recent academic papers on a particular topic. I would ask it to create a table consisting of the papers: title, year, summary, DOI or similar URL. This allowed me to leverage the LLM's speed for searching the internet like a more powerful keyword search.

I sometimes used Google Scholar to search for particular keywords in paper titles.

2.2 Skin Lesion Classification Models

What is a skin lesion classification model?
Type of Machine Learning model trained to identify lesion areas in images of skin areas [3, 4].
Choosing an appropriate pre-existing model or training my own will determine the baseline performance I will evaluate against later on.

Existing models: Multi-modal approach on an FPGA which combines image features with patient metadata [3]. Uses MobileNetV2 architecture.
ADD PAPERS FOUND FROM SURVEY REVIEW HERE

Architectures:
Fundamental MobileNet paper with X features [2].

2.3 Machine Learning Models on FPGA

Very popular fundamental paper presenting MobileNet architecture [2]

2.4 Compressing Machine Learning Models to FPGA

Papers on useful compression techniques for certain model types

2.5 Further Research Recommendations

Identify gaps, need to repeat any research

2.6 Acknowledgements

Sources for literature/statements
Continuing idea by supervisor Tess Watt in her paper but deployed to FPGA instead of RPi

[4]
Survey of recent methods for skin lesion classification as of 2023 [1]

3 Methodology

Detailed description of necessary methods, requirements, research questions that will be employed/ needed to complete the project

3.1 Requirements

MOSCOW
Enumerated to be easier to reference again

3.2 Plan for Execution

Explain how I plan to use experiments to fulfil all the requirements & objectives

3.3 Experiment Planning

How I plan to conduct experiment(s) and what research question does it address/answer (detailed)

4 Feasibility

4.1 Preliminary Work

Summary of work carried out to prepare for this project to validate feasibility.

4.2 Evaluation Strategy

Identify for the project:
• What needs evaluating
• Why it should be evaluated
• How it should be evaluated

5 Conclusions

Short summary of this doc

References

[1] Taye Girma Debelee. 2023. Skin Lesion Classification and Detection Using Machine Learning Techniques: A Systematic Review. *Diagnostics* 13, 19 (2023). doi:10.3390/diagnostics13193147

[2] Andrew G. Howard, Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam. 2017. MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications. arXiv:1704.04861 [cs.CV] https://arxiv.org/abs/1704.04861

[3] Zhihao Mao, Jin Han, and Mingzhu Shi. 2025. Design and Implementation of a Real-Time Embedded System for Early Skin Cancer Detection. *Journal of Engineering and Science in Medical Diagnostics and Therapy* 9, 2 (12 2025), 021202. arXiv:https://asmedigitalcollection.asme.org/medicaldiagnostics/article-pdf/9/2/021202/7529299/jesmdt-25-1029.pdf doi:10.1115/1.4069441

[4] Tess Watt, Christos Chrysoulas, Peter J. Barclay, Brahim El Boudani, and Grigorios Kalliatakis. 2024. Moving Healthcare AI Support Systems for Visually Detectable Diseases to Constrained Devices. *Applied Sciences* 14, 24 (2024). doi:10.3390/app142411474

6 Project Management

6.1 Gantt Chart

Plan of activities to be carried out in semester 1, then future plan for semester 2

6.2 Analysis of Risks

Including mitigation plans

6.3 PLES

A well-researched consideration of any Professional, Legal, Ethical, and Social Issues pertinent to the project. (e.g. codes of conduct (BCS), codes of practice, standards, computer law, ethical decision making, intellectual property, social aspects, copyright, data protection, and so on).

7 Gen AI Summary

Summarising use of GenAI in project/deliverables